

Cybersecurity Incident Report:

Analyzing network Attacks

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is a SYN Flood attack (a type of Denial of Service - DoS attack).

The logs show that the web server is being overwhelmed by a large number of TCP SYN requests from an unfamiliar IP address.

This event could be a SYN Flood attack where the attacker is sending numerous SYN packets without completing the TCP handshake, exhausting the server's resources.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. The three steps of the handshake are:

1. The client sends a **SYN** packet to the server requesting to establish a connection.
2. The server responds with a **SYN-ACK** packet, acknowledging the request and reserving resources for the connection.
3. The client sends a final **ACK** packet to complete the handshake and establish the connection.

The attacker sends a massive volume of SYN packets (often with spoofed IP addresses) but does not send the final ACK packets. This causes the server to keep many half-open connections in its queue, quickly exhausting available memory and resources.

The logs show an unusually high number of incoming SYN requests from a single unfamiliar IP with very few completed handshakes. As a result, the server's connection backlog fills up, preventing it from accepting new legitimate connections. This leads to the web server becoming unresponsive and users receiving connection timeout errors.